

where only the odd values of n are considered in the summation because the terms for the even values of n vanish.

2-12 SPLITTING UP OF NONHOMOGENEOUS PROBLEMS INTO SIMPLER PROBLEMS

When the heat conduction problem is nonhomogeneous due to the nonhomogeneity of the differential equation and/or the boundary conditions, it can be split up into a set of simpler problems that may be solved by the method of the separation of variables. Here we consider a nonhomogeneous problem in which the generation term and the nonhomogeneous parts of the boundary-condition functions *do not depend on time*:

$$\nabla^2 T(\mathbf{r}, t) + \frac{1}{k} g(\mathbf{r}) = \frac{1}{\alpha} \frac{\partial T(\mathbf{r}, t)}{\partial t} \quad \text{in region } R, \quad t > 0 \quad (2-145a)$$

$$k_i \frac{\partial T}{\partial \eta_i} + h_i T = f_i(\mathbf{r}) \quad \text{on boundary } S_i, \quad t > 0 \quad (2-145b)$$

$$T = F(\mathbf{r}) \quad \text{for } t = 0, \text{ in region } R \quad (2-145c)$$

where $(\partial/\partial \eta_i)$ = derivative along the outward-drawn normal to the boundary surface S_i ($i = 1, 2, \dots, N$) and N = number of continuous boundary surfaces of region R . Here we note that $g(\mathbf{r})$ and $f_i(\mathbf{r})$ *do not depend on time*. Clearly, many special cases are obtainable from the general problem given above. We shall now split up this problem into a number of simpler problems in the following manner:

1. A set of steady-state problems defined by the temperatures $T_{0j}(\mathbf{r})$, $j = 0, 1, 2, \dots, N$.
2. A homogeneous time-dependent problem defined by the temperature $T_h(\mathbf{r}, t)$.

The temperatures $T_{0j}(\mathbf{r})$ are taken as the solutions of the following set of steady-state problems

$$\nabla^2 T_{0j}(\mathbf{r}) + \delta_{0j} \frac{1}{k} g(\mathbf{r}) = 0 \quad \text{in region } R \quad (2-146a)$$

$$k_i \frac{\partial T_{0j}}{\partial \eta_i} + h_i T_{0j} = \delta_{ij} f_i(\mathbf{r}) \quad \text{on boundary } S_i \quad (2-146b)$$

where

$$i = 1, 2, \dots, N$$

$$j = 0, 1, 2, \dots, N$$

N = number of continuous boundary surfaces of region R
 δ_{ij} = Kronecker delta = $\begin{cases} 0 & \text{for } i \neq j \\ 1 & \text{for } i = j \end{cases}$

The temperature $T_h(\mathbf{r}, t)$ is taken as the solution of the following homogeneous problem:

$$\nabla^2 T_h(\mathbf{r}, t) = \frac{1}{\alpha} \frac{\partial T_h(\mathbf{r}, t)}{\partial t} \quad \text{in region } R, \quad t > 0 \quad (2-147a)$$

$$k_i \frac{\partial T_h}{\partial \eta_i} + h_i T_h = 0 \quad \text{on boundary } S_i \quad (2-147b)$$

$$T_h = F(\mathbf{r}) - \sum_{j=0}^N T_{0j}(\mathbf{r}) \quad \text{for } t = 0, \text{ in region } R \quad (2-147c)$$

Then, the solution $T(\mathbf{r}, t)$ of the problem (2-145) is given in terms of the solutions of the above problems as

$$T(\mathbf{r}, t) = T_h(\mathbf{r}, t) + \sum_{j=0}^N T_{0j}(\mathbf{r}) \quad (2-148)$$

The validity of equation (2-148) can be verified by substituting this equation into equation (2-145) and by utilizing equations (2-146) and (2-147).

We note that equations (2-146) corresponds to a set of steady-state heat conduction problems. The function $T_{00}(\mathbf{r})$ for $j = 0$ corresponds to a steady-state heat conduction problem with heat generation in the medium, but subject to all homogeneous boundary conditions. The functions $T_{01}(\mathbf{r})$, $T_{02}(\mathbf{r})$, $T_{03}(\mathbf{r})$, ... for $j = 1, 2, 3, \dots$, respectively, corresponds to heat conduction problems with no heat generation, but only one of the boundary conditions, $i = j$, is nonhomogeneous.

The homogeneous problem given by equations (2-147) is the homogeneous version of the original problem (2-145), except the initial condition is modified by subtracting from it the sum of the solutions of the steady-state problems (2-146).

Clearly, the problems defined by equations (2-146) and (2-147), when given in the rectangular coordinate system, are soluble with the techniques discussed in this chapter. The more general case will be discussed in Chapter 13 in connection with the general method of solution of heat-conduction problems by the integral transform technique.

Example 2-15

A slab, $0 \leq x \leq L$, is initially at temperature $F(x)$. For times $t > 0$ the boundaries at $x = 0$ and $x = L$ are kept at constant temperatures T_1 and T_2 , respectively. Obtain an expression for the temperature distribution $T(x, t)$ in the slab.

Solution. The mathematical formulation of this problem is given as

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{in} \quad 0 < x < L, \quad t > 0 \quad (2-149a)$$

$$T = T_1 \quad \text{at} \quad x = 0, \quad t > 0 \quad (2-149b)$$

$$T = T_2 \quad \text{at} \quad x = L, \quad t > 0 \quad (2-149c)$$

$$T = F(x) \quad \text{for} \quad t = 0, \quad \text{in the region} \quad (2-149d)$$

Since the problem is one-dimensional, we split it into a steady-state problem for $T_s(x)$ given as

$$\frac{d^2 T_s}{dx^2} = 0 \quad \text{in} \quad 0 < x < L \quad (2-150a)$$

$$T_s = T_1 \quad \text{at} \quad x = 0 \quad (2-150b)$$

$$T_s = T_2 \quad \text{at} \quad x = L \quad (2-150c)$$

and to a homogeneous problem for $T_h(x, t)$ given by

$$\frac{\partial^2 T_h}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T_h}{\partial t} \quad \text{in} \quad 0 < x < L, \quad t > 0 \quad (2-151a)$$

$$T_h = 0 \quad \text{at} \quad x = 0 \quad \text{and} \quad x = L, \quad t > 0 \quad (2-151b)$$

$$T_h = F(x) - T_s(x) \equiv f^*(x) \quad \text{for} \quad t = 0, \quad \text{in the region} \quad (2-151c)$$

Then, the solution for the original problem (2-149) is determined from

$$T(x, t) = T_s(x) + T_h(x, t) \quad (2-152)$$

The solution of the steady-state problem (2-150) is given as

$$T_s(x) = T_1 + (T_2 - T_1) \frac{x}{L} \quad (2-153)$$

The solution of the problem (2-151) is immediately written from equation (2-36a) as

$$T_h(x, t) = \sum_{m=1}^{\infty} e^{-\alpha \beta_m^2 t} \frac{1}{N(\beta_m)} X(\beta_m x) \int_0^L X(\beta_m x') f^*(x') dx' \quad (2-154)$$

where the eigenfunctions $X(\beta_m x)$, the norm $N(\beta_m)$ and the eigenvalues β_m are obtained from Table 2-2, case 9, as

$$X(\beta_m, x) = \sin \beta_m x, \quad \frac{1}{N(\beta_m)} = \frac{2}{L} \quad (2-155a)$$

and the β_m values are the roots of

$$\sin \beta_m L = 0 \quad (2-155b)$$

and the initial condition function $f^*(x)$ is defined as

$$f^*(x) \equiv F(x) - T_s(x) = F(x) - T_1 - (T_2 - T_1) \frac{x}{L} \quad (2-155c)$$

The solution $T(x, t)$ of the problem (2-149) is obtained by introducing equations (2-153) and (2-154) into equation (2-152). We find

$$T(x, t) = T_1 + (T_2 - T_1) \frac{x}{L} + \sum_{m=1}^{\infty} e^{-\alpha \beta_m^2 t} \sin \beta_m x \cdot \int_0^L [F(x') - T_1 - (T_2 - T_1) \frac{x'}{L}] \sin \beta_m x' dx' \quad (2-156a)$$

Performing the integrations we obtain

$$T(x, t) = T_1 + (T_2 - T_1) \frac{x}{L} + \frac{2}{L} \sum_{m=1}^{\infty} e^{-\alpha \beta_m^2 t} \sin \beta_m x \int_0^L F(x') \sin \beta_m x' dx' + \frac{2}{L} \sum_{m=1}^{\infty} e^{-\alpha \beta_m^2 t} \frac{1}{\beta_m} \sin \beta_m x [T_2 \cos m\pi - T_1] \quad (2-156b)$$

where $\cos m\pi = (-1)^m$ and $\beta_m = m\pi/L$.

Example 2-16

A slab, $0 \leq x \leq L$, is initially at temperature $F(x)$. For times $t > 0$, heat is generated in the solid at a constant rate of g_0 per unit volume, the boundary at $x = 0$ is kept insulated and the boundary at $x = L$ is kept at zero temperature. Obtain an expression for the temperature distribution $T(x, t)$ in the slab.

Solution. The mathematical formulation of this problem is given as

$$\frac{\partial^2 T}{\partial x^2} + \frac{1}{k} g_0 = \frac{1}{\alpha} \frac{\partial T}{\partial t} \quad \text{in} \quad 0 < x < L, \quad t > 0 \quad (2-157a)$$

$$\frac{\partial T}{\partial x} = 0 \quad \text{at} \quad x = 0, \quad t > 0 \quad (2-157b)$$

$$T = 0 \quad \text{at} \quad x = L, \quad t > 0 \quad (2-157c)$$

$$T = F(x) \quad \text{for} \quad t = 0, \quad \text{in } 0 \leq x \leq L \quad (2-157d)$$

This problem is split up into a steady-state problem for $T_s(x)$ as

$$\frac{d^2 T_s}{dx^2} + \frac{1}{k} q_0 = 0 \quad \text{in} \quad 0 < x < L \quad (2-158a)$$

$$\frac{dT_s}{dx} = 0 \quad \text{at} \quad x = 0 \quad \text{and} \quad T_s = 0 \quad \text{at} \quad x = L \quad (2-158b)$$

and a homogeneous problem for $T_h(x, t)$ as

$$\frac{\partial^2 T_h}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T_h}{\partial t} \quad \text{in} \quad 0 < x < L, \quad t > 0 \quad (2-159a)$$

$$\frac{\partial T_h}{\partial x} = 0 \quad \text{at} \quad x = 0, \quad T_h = 0 \quad \text{at} \quad x = L, \quad \text{for } t > 0 \quad (2-159b)$$

$$T_h = F(x) - T_s(x) \equiv f^*(x) \quad \text{for} \quad t = 0, \text{ in } 0 \leq x \leq L \quad (2-159c)$$

Then, the solution of the original problem (2-157) is determined from

$$T(x, t) = T_s(x) + T_h(x, t) \quad (2-160)$$

The solution of the steady-state problem (2-158) is

$$T_s(x) = \frac{1}{2k} g_0 L^2 \left(1 - \frac{x^2}{L^2} \right) \quad (2-161)$$

and the solution of the homogeneous problem (2-159) is obtained as

$$T_h(x, t) = \sum_{m=0}^{\infty} e^{-\alpha \beta_m^2 t} \cos \beta_m x \int_0^L f^*(x') \cos \beta_m x' dx' \quad (2-162a)$$

where

$$f^*(x) \equiv F(x) - \frac{1}{2k} g_0 L^2 \left(1 - \frac{x^2}{L^2} \right) \quad (2-162b)$$

and the β_m values are the positive roots of

$$\cos \beta_m L = 0 \quad \text{or} \quad \beta_m = \frac{(2m+1)\pi}{2L}, \quad m = 0, 1, 2, \dots \quad (2-162c)$$

Introducing equations (2-161) and (2-162) into (2-160) and performing the integrations we obtain

$$T(x, t) = \frac{g_0 L^2}{2k} \left(1 - \frac{x^2}{L^2} \right) + \sum_{m=0}^{\infty} e^{-\alpha \beta_m^2 t} \cos \beta_m x \int_0^L f(x') \cos \beta_m x' dx' - \frac{2g_0}{kL} \sum_{m=0}^{\infty} (-1)^m e^{-\alpha \beta_m^2 t} \frac{1}{\beta_m^3} \cos \beta_m x \quad (2-163)$$

2-13 USEFUL TRANSFORMATIONS

In this section we present some transformations that are useful in reducing the differential equation into a more convenient form.

1. We consider an equation containing convective and generation terms in the form

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} - \beta \frac{\partial T}{\partial x} + \gamma T + g \quad (2-164)$$

where α , β , and γ are constants, $\beta(\partial T/\partial x)$ represents a convective term and γT represents generation proportional to the local temperature. We define a new dependent variable $W(x, t)$ as

$$T(x, t) = W(x, t) \exp \left[\frac{\beta}{2\alpha} x - \left(\frac{\beta^2}{4\alpha} - \gamma \right) t \right] \quad (2-165)$$

Then, under this transformation, equation (2-164) reduces to

$$\frac{\partial W}{\partial t} = \alpha \frac{\partial^2 W}{\partial x^2} + g \cdot \exp \left\{ - \left[\frac{\beta}{2\alpha} x - \left(\frac{\beta^2}{4\alpha} - \gamma \right) t \right] \right\} \quad (2-166)$$

which is easier to solve than equation (2-164). The boundary and the initial conditions for the problem should be transformed with the same transformation.

2. We now generalize the above procedure to three-dimensional equation given as

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \beta_1 \frac{\partial T}{\partial x} - \beta_2 \frac{\partial T}{\partial y} - \beta_3 \frac{\partial T}{\partial z} + \gamma T + g \quad (2-167)$$

where $\alpha, \beta_1, \beta_2, \beta_3$, and γ are constants. We define a new dependent variable $W(x, y, z, t)$ as

$$T(x, y, z) = W(x, y, z, t) \exp \left[\frac{\beta_1}{2\alpha} x - \left(\frac{\beta_2}{4\alpha} - \gamma \right) t \right] \cdot \exp \left[\frac{\beta_2}{2\alpha} y + \frac{\beta_3}{4\alpha} t \right] \quad (2-168)$$

Under this transformation equation (2-167) reduces to

$$\frac{\partial W}{\partial t} = \alpha \left(\frac{\partial^2 W}{\partial x^2} + \frac{\partial^2 W}{\partial y^2} + \frac{\partial^2 W}{\partial z^2} \right) + G \quad (2-169a)$$

where

$$G = g \exp \left[-\frac{\beta_1}{2\alpha} x + \left(\frac{\beta_2}{4\alpha} - \gamma \right) t \right] \cdot \exp \left[-\frac{\beta_2}{2\alpha} y + \frac{\beta_3}{4\alpha} t \right] \cdot \exp \left[-\frac{\beta_3}{2\alpha} z + \frac{\beta_3}{4\alpha} t \right] \quad (2-169b)$$

which is easier to solve than equation (2-167).

2-14 TRANSIENT-TEMPERATURE CHARTS

Temperature-time charts are useful for rapid estimation of temperature history in solids, and for some specific situations such charts can be prepared. Here we consider a plate of thickness $2L$, initially at a uniform temperature T_i and for times $t > 0$ it is subjected to convection from both its surfaces into an ambient at a constant temperature T_∞ with a heat transfer coefficient h . Because of symmetry, we choose the origin of the x coordinate at the center of the plate and consider only half of the plate. The mathematical formulation of this transient heat conduction problem is given in the dimensionless form as

$$\frac{\partial^2 \theta}{\partial X^2} = \frac{\partial \theta}{\partial \tau} \quad \text{in} \quad 0 < X < 1, \quad \text{for} \quad \tau = 0 \quad (2-170a)$$

$$\frac{\partial \theta}{\partial X} = 0 \quad \text{at} \quad X = 0, \quad \text{for} \quad \tau > 0 \quad (2-170b)$$

$$\frac{\partial \theta}{\partial X} + \text{Bi} \theta = 0 \quad \text{at} \quad X = 1, \quad \text{for} \quad \tau > 0 \quad (2-170c)$$

$$\theta = 1 \quad \text{in} \quad 0 \leq X \leq 1, \quad \text{for} \quad \tau = 0 \quad (2-170d)$$

where the following dimensionless parameters are introduced

$$\theta = \frac{T(x, t) - T_\infty}{T_i - T_\infty} = \text{dimensionless temperature} \quad (2-171a)$$

$$X = \frac{x}{L} = \text{dimensionless coordinate} \quad (2-171b)$$

$$\text{Bi} = \frac{hL}{k} = \text{Biot number} \quad (2-171c)$$

$$\tau = \frac{\alpha t}{L^2} = \text{dimensionless time or Fourier number} \quad (2-171d)$$

It is instructive to examine the physical significance of the dimensionless parameters τ and Bi.

The dimensionless time τ is rearranged in the form

$$\tau = \frac{\alpha t}{L^2} = \frac{\left(\frac{\text{rate of heat conduction}}{\text{across } L \text{ in volume } L^3, \text{ W/}^\circ\text{C}} \right)}{\left(\frac{\text{rate of heat storage}}{\text{in volume } L^3, \text{ W/}^\circ\text{C}} \right)}$$

Thus, the Fourier number is a measure of the rate of heat conduction compared with the rate of heat storage in a given volume element. Therefore, the larger the Fourier number, the deeper the penetration of heat into a solid over a given time.

The physical significance of the Biot number has already been discussed in the previous chapter in connection with the lumped analysis. It represents the ratio of the "internal thermal resistance" to the "external thermal resistance."

The solution of the transient heat conduction problem (2-170) is presented in the graphical form in Fig. 2-13 *a, b*. Here, Fig. 2-13 *a* gives the midplane temperature T_0 or $\theta(0, \tau)$ at $X = 0$ as a function of the dimensionless time τ for several different values of the parameter $1/\text{Bi}$. The curve for $1/\text{Bi} = 0$ corresponds to the case in which $h \rightarrow \infty$, or the surfaces of the plate are maintained at the ambient temperature T_∞ . For large values of $1/\text{Bi}$, the Biot number is small, or the internal conductance of the solid is large in comparison with the heat transfer coefficient at the surface. This, in turn, implies that the temperature distribution within the solid is sufficiently uniform, and hence lumped system analysis becomes applicable.

Figure 2-13 *b* relates the temperature at six different locations within the slab to the midplane temperature T_0 [i.e., $\theta(0, \tau)$]. Thus, given T_0 , temperature at these locations can be determined. An examination of Fig. 2-13 *b* reveals that for values of $1/\text{Bi}$ larger than 10, or $\text{Bi} < 0.1$, the temperature distribution within the slab may be considered uniform with an error of less than about 5%; hence for such